

VANCOUVER PEARSON, WA, USA

WMO: 727918

Lat: 45.621N

Lon: 122.657W

Elev: 9

StdP: 101.22

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94298

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.2	-2.6	-12.8	1.2	0.0	-8.8	1.8	0.8	9.2	5.0	8.2	5.8	1.2	140	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	33.0	19.3	31.0	18.9	28.7	18.2	20.6	30.3	19.7	29.0	18.8	27.4	3.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	23.7	16.2	11.5	22.7	15.3	10.9	22.0	59.1	30.2	56.0	28.8	53.1	27.6	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	5.8	5.2	-7.6	37.3	2.8	1.8	-9.7	38.5	-11.3	39.6	-12.9	40.5	-14.9	41.8		
(5)			-8.3	22.2	2.8	1.2	-10.3	23.1	-12.0	23.8	-13.5	24.4	-15.6	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	5.0	6.2	8.5	10.9	14.5	17.5	20.6	20.9	17.8	12.6	8.1	4.9	(6)
(7)		DBStd	6.43	3.38	2.83	2.69	2.88	3.25	3.16	2.75	2.65	2.94	2.73	3.41	3.50	(7)
(8)		HDD10.0	594	158	108	61	21	2	0	0	0	0	8	74	162	(8)
(9)		HDD18.3	2422	414	339	304	223	127	53	8	7	45	178	307	417	(9)
(10)		CDD10.0	1445	2	2	16	49	142	224	329	338	234	89	17	3	(10)
(11)		CDD18.3	229	0	0	0	1	8	27	79	86	29	1	0	0	(11)
(12)		CDH23.3	2526	0	0	0	27	154	334	800	838	360	12	0	0	(12)
(13)		CDH26.7	960	0	0	0	5	49	128	312	342	123	1	0	0	(13)
(14)	Wind	WSAvg	2.1	2.1	2.2	2.1	2.0	2.2	2.4	2.1	1.8	1.7	2.0	2.3	(14)	
(15)	Precipitation	PrecAvg	1037	154	112	103	78	64	42	18	27	45	85	154	169	(15)
(16)		PrecMax	1637	326	269	182	196	157	102	95	130	122	187	328	337	(16)
(17)		PrecMin	708	7	21	22	24	4	1	0	0	0	6	29	42	(17)
(18)		PrecStd	192	72	56	39	35	32	25	19	28	32	45	73	71	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.0	21.1	27.0	31.2	34.5	36.1	33.1	25.1	18.2	15.1	(19)	
(20)			MCWB	11.6	9.3	12.7	15.9	17.4	19.8	20.4	20.2	18.7	15.4	14.3	13.1	(20)
(21)		2%	DB	12.5	13.8	17.9	22.6	27.6	30.2	32.5	32.9	30.1	22.1	16.1	12.5	(21)
(22)			MCWB	10.0	9.0	11.0	13.6	16.2	18.7	19.6	19.4	17.4	14.1	13.1	10.5	(22)
(23)		5%	DB	11.3	12.4	15.7	19.1	24.1	27.3	30.1	30.5	27.5	19.8	14.3	11.1	(23)
(24)			MCWB	9.1	8.5	10.0	11.7	15.0	17.4	19.2	19.2	17.2	13.6	11.6	9.2	(24)
(25)		10%	DB	9.8	11.0	13.6	17.0	21.4	24.1	27.7	27.9	24.9	17.6	12.9	9.5	(25)
(26)			MCWB	7.8	7.9	9.5	11.1	14.0	16.0	18.3	18.4	16.4	13.1	10.5	7.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	11.7	13.9	16.6	18.9	21.1	22.2	21.7	20.6	17.6	15.6	13.3	(27)
(28)			MCD B	14.0	13.4	19.1	25.3	29.3	32.1	32.2	32.8	29.6	21.8	17.2	15.0	(28)
(29)		2%	WB	10.6	10.4	12.1	14.3	17.1	19.2	20.5	20.6	19.0	15.8	13.5	10.9	(29)
(30)			MCD B	11.9	12.6	15.6	20.9	25.6	28.5	29.9	30.4	27.0	19.5	15.4	12.3	(30)
(31)		5%	WB	9.2	9.2	10.9	12.8	15.8	17.8	19.6	19.7	17.9	14.8	12.1	9.3	(31)
(32)			MCD B	10.8	11.4	14.2	17.8	22.7	25.7	29.0	28.8	25.2	18.0	13.8	10.6	(32)
(33)		10%	WB	8.0	8.1	10.1	11.6	14.6	16.6	18.6	18.7	17.0	13.8	11.1	7.9	(33)
(34)			MCD B	9.4	10.2	13.3	15.5	20.3	23.3	27.2	27.1	23.3	16.8	12.7	9.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	7.9	8.7	9.6	10.4	10.6	12.4	12.7	12.2	9.4	6.9	5.8	(35)
(36)			MCD BR	7.4	10.5	13.2	16.0	17.3	16.9	17.1	17.7	17.8	14.1	7.9	6.5	(36)
(37)			MCWBR	5.4	6.4	7.5	8.2	8.0	7.2	6.8	6.8	7.3	7.3	5.2	5.1	(37)
(38)		5% WB	MCD BR	6.8	7.8	10.4	13.7	14.6	15.1	15.3	15.6	14.9	9.8	6.1	5.9	(38)
(39)			MCWBR	5.4	5.3	6.5	7.5	7.1	6.8	6.7	6.4	6.6	5.8	4.6	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.317	0.330	0.361	0.377	0.362	0.350	0.357	0.343	0.334	0.322	0.326	(40)	
(41)		taud	2.532	2.553	2.524	2.403	2.375	2.431	2.491	2.486	2.506	2.545	2.538	2.476	(41)	
(42)		Ebn at Noon	793	873	905	899	891	904	911	893	878	838	782	737	(42)	
(43)		Edh at Noon	66	77	90	111	118	112	104	101	91	76	65	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.15	1.99	2.90	4.23	5.38	5.95	6.76	5.78	4.32	2.50	1.35	0.94	(44)	
(45)		RadStd	0.19	0.28	0.38	0.40	0.52	0.56	0.48	0.39	0.38	0.31	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+79
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+26

Nomenclature: See separate page